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Advanced Control System Design - Tutorial 6 – Solution

1. Investigate the observability of the following system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-8 & -17 & -10\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}u$$

$$y=\begin{bmatrix}0 & 6 & 6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Solution:

Using the standard formula:

$$\mathbf{O}=\begin{bmatrix}\mathbf{C}^T & \mathbf{A}^T\mathbf{C}^T & (\mathbf{A}^T)^2\mathbf{C}^T\end{bmatrix}=\begin{bmatrix}0 & 6 & -42\\ 6 & 6 & -102\\ 6 & 0 & -60\end{bmatrix}\Rightarrow|\mathbf{O}|=0$$

The system is not completely observable.

Using the transfer function approach:

$$G(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+D$$

$$G(s)=\begin{bmatrix}0 & 6 & 6\end{bmatrix}\begin{bmatrix}s+8 & 17 & 10\\ -1 & s & 0\\ 0 & -1 & s\end{bmatrix}^{-1}\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}+0$$

$$G(s)=\begin{bmatrix}0 & 6 & 6\end{bmatrix}\left[\frac{1}{s^3+8s^2+17s+10}\begin{bmatrix}s^2 & -17s-10 & -10s\\ s & s(s+8) & -10\\ 1 & s+8 & s^2+8s+17\end{bmatrix}\right]\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$$

$$G(s)=\frac{6s+6}{s^3+8s^2+17s+10}=\frac{6\cancel{(s+1)}}{\cancel{(s+1)}(s+2)(s+5)}$$

Not completely observable due to the zero-pole cancellation

Using inspection (must be transformed to the diagonal or Jordan form first):

Matrix **A** is not in the required form for diagonalization (refer to Lecture 4 slides). Thus, the system must be transformed into the CCF (refer to Lecture 3 slides):

Based on the previous transfer function (before cancellation), we get:

$$\mathbf{A}_{CCF}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -10 & -17 & -8\end{bmatrix},\ \mathbf{B}_{CCF}=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix},\ \mathbf{C}_{CCF}=\begin{bmatrix}6 & 6 & 0\end{bmatrix},\ \mathbf{D}_{CCF}=0$$

$$\lambda_1=-1,\ \lambda_2=-2,\ \lambda_3=-5$$

$$\mathbf{P}=\begin{bmatrix}1 & 1 & 1\\ \lambda_1 & \lambda_2 & \lambda_3\\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2\end{bmatrix}=\begin{bmatrix}1 & 1 & 1\\ -1 & -2 & -5\\ 1 & 4 & 25\end{bmatrix}$$

$$\dot{\mathbf{z}}=\mathbf{P}^{-1}\mathbf{A}_{CCF}\mathbf{P}\mathbf{z}+\mathbf{P}^{-1}\mathbf{B}_{CCF}u$$

$$\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\\ \dot{z}_3\end{bmatrix}=\begin{bmatrix}2.500 & 1.750 & 0.250\\ -1.667 & -2.000 & -0.333\\ 0.167 & 0.250 & 0.083\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -10 & -17 & -8\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ -1 & -2 & -5\\ 1 & 4 & 25\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}$$
$$+\begin{bmatrix}2.500 & 1.750 & 0.250\\ -1.667 & -2.000 & -0.333\\ 0.167 & 0.250 & 0.083\end{bmatrix}\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\\ \dot{z}_3\end{bmatrix}=\begin{bmatrix}-1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & -5\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}+\begin{bmatrix}0.250\\ -0.333\\ 0.083\end{bmatrix}u$$

$$y=\mathbf{C}_{CCF}\mathbf{P}\mathbf{z}=\begin{bmatrix}6 & 6 & 0\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ -1 & -2 & -5\\ 1 & 4 & 25\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}=\begin{bmatrix}0 & -6 & -24\end{bmatrix}$$

The system is not completely observable.